

CMPT 280

Topic 19: Graphs

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References

- Textbook, Chapter 19

Graphs

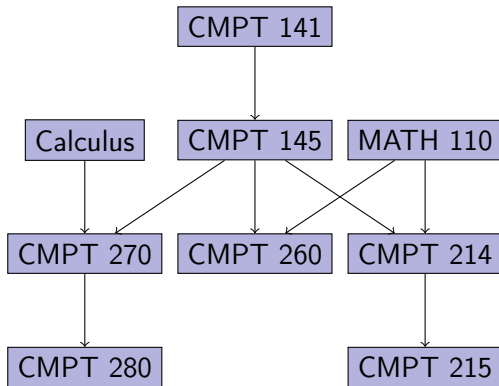
Definition

A *graph* $G = (V, E)$ consists of a set V of *vertices* (or *nodes*) and a set $E \subseteq V \times V$ of *edges* (or *arcs*).

A graph is a representation of a mathematical relation. The actual relation depends on what the vertices represent.

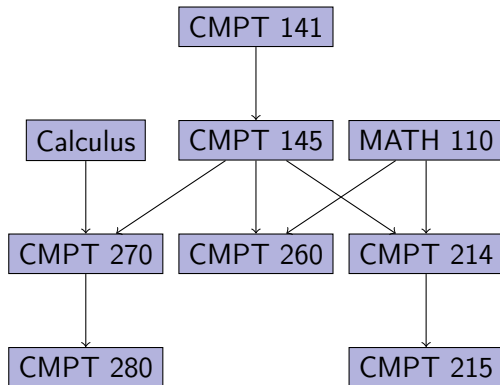
Graphs as Relations

If graph vertices represent courses, then $E \subseteq V \times V$ could be the relation "is a prerequisite for".



In what ways is this **not** a tree?

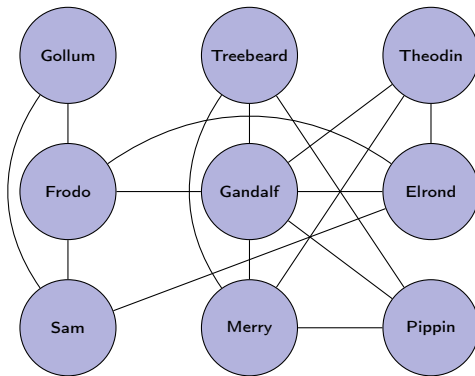
Exercise 1



- a) What are the sets V and E for this graph?
- b) What is the indegree of CMPT 214? CMPT 280?
- c) What is the outdegree of CMPT 145? CMPT Math 110?

Exercise 2

Subgraph of Social Graph of "Return of the King"



Source: <http://visualizing.org/full-screen/112639> (link now dead)

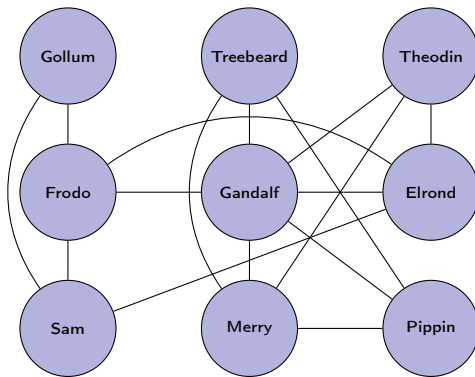
a) What is the degree of "Gandalf"? "Gollum"?

Exercise 2 (continued)

- b) Give examples of a walk from "Treebeard" to "Gollum".
- c) Give examples of a trail from "Theodin" to "Sam".
- d) Give examples of a path from "Merry" to "Frodo".
- e) What is the shortest path from "Gollum" to "Pippin"?
- f) Give an example of a *Hamiltonian Path* in this graph.
A Hamiltonian Path is a path that visits every vertex.
- g) Give an example of a circuit that starts and ends at Gandalf.
- h) Give an example of a cycle that starts and ends at Gandalf.
- i) How many connected components does this graph have?
- j) If we removed "Frodo" and "Eldrond", how many connected components would the graph have?

Exercise 3

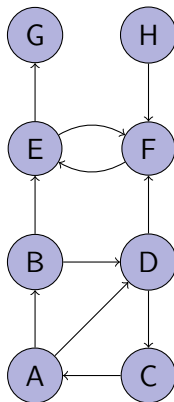
Subgraph of Social Graph of "Return of the King"



Source: <http://visualizing.org/full-screen/112639>

Give an example of a subgraph of the above graph that only includes hobbits (i.e. Frodo, Sam, Merry, and Pippin).

Exercise 4



Assume that each node is connected to itself with a self-loop (not shown).

What are the strongly connected components of the above graph?

Next Class

- **Next class reading:** Chapter 20: Graph Data Structures